

Paper Directions 25 — my read (Aug 21, 2026)

Source: `transcripts/20260821_Paper_Directions_25_otter_ai_transcript.docx` (~21 min)
Whiteboard: `transcripts/PD25_board.jpeg`
Context: Alex returned after thinking about `fix Â_i`; Charles had been building minutes BDP (APGMS/ARGMS) and porch CCT read.
Prior: `PD23_notes.md` (MLE next) · `PD24_notes.md` (BPM extra) · `../3-Master_Plan/re_entry/SCOUT_and_COMPASS/CCT_Campaign_Plan.md`

Headline

Alex’s **canonical empirical test** for fixed $\hat{A}$: plot **P(success)** vs **team talent** and expect a **flat plateau**, then a **congestion downturn** only when team quality pushes enough peers past the selection threshold (K/N , linked to θ and γ in the model). **Bin the X-axis on purpose** — need $\geq$ **two bins after** the congestion zone or you will never see the downturn. **Explore empirically first** (CCT), then connect inflection to the generative model.

Whiteboard (PD25_board.jpeg)

Element	Read
Title	Fix A_i group (top performers / top log scale)
Main axes	X = team talent (average team $\hat{A}$ / T_j) · Y = P(s) = draft rate
Small sketch	Ability distribution for fixed group; K/N marked on upper tail
Main curve	Flat , then sharp drop at congestion onset
Formula at knee	(K/N – θ) / γ — inflection is selection-mechanism dependent
Bracket	High team-talent region $\geq K/N$ (“Team A_i”)

This is **not** the marginal hero curve. It is **conditional on $\hat{A}$** with **team environment on X**.

Alex — key quotes (paraphrased)

- Shape**: “For a fixed AI group ... flat ... then congestion kicks in ... downward trend.”
- Where**: Downturn should appear only when team talent gets **close to K/N** (NCAA: **very high** on X because K is tiny).
- Story**: “All the action happens **up here**” — zoom the high team-talent tail, not league-wide marginals.
- Binning**: Optimize **X resolution** — need **at least two points after** the downturn; blind 16-quantile ventiles may miss it. Back-of-envelope: “I’m never going to expect this with **under 20 bins**” — use calculation to **prune sweeps**, not replace them.
- Power tradeoff**: More bins + finer $\hat{A}$ slices $\rightarrow$ fewer people per cell; **K is very small** in NBA draft data.
- Order**: “Almost say just first explore **empirically** still” — then fit inflection in the model.
- Minutes**: Charles’s APGMS/ARGMS / min-floor work — Alex **yes** (panel who-counts); separate from but feeds estimand choice.

How this maps to CCT work (Aug 21)

Alex ask	Already built	Gap
Fix $\hat{A}$, conditional draft rate	P1 matched band $\times$ poolq_loo (+DFT whisper ~2 pp)	✓ partial win on peer axis
Same, **T_j on X**	P2b tail bins — flat-then-down (+DFT)	✓ YES — plateau ~34% $\rightarrow$ tail ~8% (z [2,3])
Empirical first	Act II sweep, BDP, minutes plots, triptych	✓ on track
K/N-aware X bins	P2b piecewise 4+20 tail bins	✓ shipped
Inflection = f(θ , γ , K/N)	Not yet	Act III / MLE bridge after P2b

Reconciliation: P1 on **poolq_loo** is the right *peer-congestion* microscope. Alex’s board is the *team-talent* version of the same fix- $\hat{A}$ logic. Both can be true: full panel ****T_j**** rises (program quality); **+DFT high tail** may still show flat-then-down if binned correctly.

Engineering ticket — Priority 2b — SHIPPED 2026-08-21

See `CCT_Campaign_Plan.md` §3 Priority 2b · outputs in `basic_data_plots/CCT_draft_rate_fixedAi_Tj_knbins_*`.

One-liner for Alex: “We’re holding $\hat{A}$ fixed and binning the **high team-talent tail** on purpose — and we see your flat-then-down picture on +DFT.”

Open / homework

- ☐ COMPASS: K/N → minimum X-bin count memo (θ , γ , roster N, draft K)
- ☒ SCOUT: P2b figure + JSON with `edge_lo` / `edge_hi` / `n` per bin (2026-08-21)
- ☐ Charles: eyeball P2b + triptych — rehearse §8 Alex paragraph for HAND
- ☐ Later: map empirical knee to model inflection (MLE phase)
- ☐ **After next week (Alex):** paper framing + outline (placeholders OK); then 50% writing / 50% code-data

Artifacts

Item	Path
Transcript	<code>transcripts/20260821_Paper_Directions_25_otter_ai_transcript.docx</code>
Whiteboard	<code>transcripts/PD25_board.jpeg</code>
Campaign addendum	<code>3-Master_Plan/re_entry/SCOUT_and_COMPASS/CCT_Campaign_Plan.md</code> §3 Priority 2b